



CFR Rookie Task - ELEC - PCB Design

Scope:

The goal of this project is to make a PCB schematic in Altium that reads the outputs of 4 different sensors into an MCU which then outputs those readings over CAN (you don't need to worry about firmware, only electrical schematics).

The 4 sensors are:

1. Brake Sensor: INFKL-800
2. Wheel Speed Sensor: LJ12A3-4-Z/BX
3. Temperature sensor/Thermistor
4. A random 0-12V analog signal with a maximum frequency of 10kHz

Requirements:

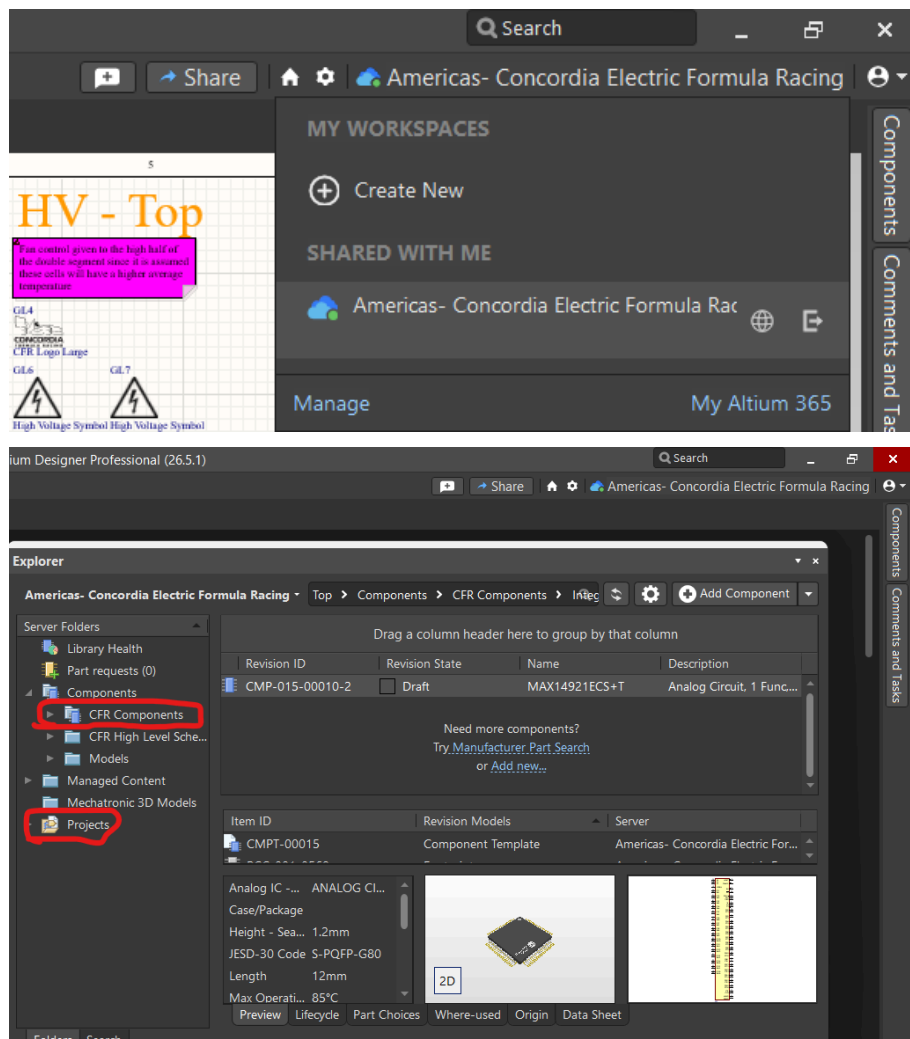
1. The input power of the board must be 12VDC (same as the car's low voltage battery)
2. An appropriately sized connector (voltage, current, # of inputs) must be used
3. The STM32f105 or 103 must be used as the MCU, explain your choice
4. The ADC resolution of each analog input should be as large as possible and the full scale of the sensors' measurement must be readable. Explain the design choices you made to do this
5. CAN Bus must have termination and pass through on this board
6. This project should only be saved locally on your computer. Do not push your project to our Altium server.

Other hints:

- It is encouraged that you explore our previous designs for common building blocks. Particularly for this project, the VCU schematic will be helpful
- For the MCU please use the existing reusable blocks we have in Altium which already contains all the necessary external components for the STM32f105/3
- CAN is a communication protocol in the physical and data link layers. It is not necessary to know everything about the protocol but to complete this project you will need to know how it works on a high level.
- Altium has lots of documentation about how to use their software

Altium Getting Started:

- Once you download and sign into Altium, you will need to begin using one of the licenses. We have a total of 10. Your license will get released once you close Altium.
- Now you can access our workspace/server “Americas - Concordia Electric Formula Racing”
- Open our projects:



- It is better to use the components already existing in our team's library when possible, under “Components->CFR Components”
- Under “Projects” is where you can find our current and past schematics and PCBs